

Report
Of
Data Science Master Virtual Internship

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
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Batch: 2026-27

CANDIDATE'S DECLARATION

I, Aditya Kumar, do hereby declare that the internship report titled “Data Science Master Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Aditya Kumar

Roll No.: 2410031133

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Aditya Kumar

Roll No.: 2410031133

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INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Aditya Kumar

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
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Certificate ID : 4369aaa8eb3c6f9653b0

Student ID: STU6998813a78ffd1771602234



GRADE- O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

This report presents the work undertaken during the Data Science Master Virtual Internship, an 8-week program conducted under the AICTE–EduSkills National Internship Portal in association with Siemens Industry Software and delivered on the RapidMiner (an Altair company) data science platform. The internship was undertaken between April 2026 and June 2026 while pursuing the Bachelor of Technology in Computer Science and Engineering at IILM University, Greater Noida.

The program followed a week-wise structured learning plan combining conceptual instruction with hands-on exercises in RapidMiner Studio and RapidMiner AI Hub. Progress was tracked through weekly assessments, project documentation and assignments, culminating in a Final Assessment Test in the last week. Upon successful completion of all assessment requirements, the internship was awarded an overall grade of “O – Outstanding” (90–100), and seven industry-recognized RapidMiner certifications were earned across the Professional and Master levels.

4.2 Organization Profile

Siemens Industry Software (India) Pvt. Ltd. is part of Siemens Digital Industries Software, a global provider of software and services for product design, manufacturing and industrial automation. As part of its skill-development initiatives, Siemens supported the Data Science Master track offered under this internship.

The internship was delivered through the National Internship Portal, an initiative of the All India Council for Technical Education (AICTE) under the Ministry of Education, Government of India, in partnership with EduSkills Foundation — a not-for-profit organization focused on building industry-relevant skills among engineering students through structured virtual internships.

The hands-on component of the internship was built entirely on RapidMiner, a data science and machine learning platform developed by Altair Engineering Inc. RapidMiner provides an end-to-end visual environment for data access, preparation, modeling, validation, deployment and platform administration, and was used throughout the internship for all exercises, projects and certification examinations.

4.3 Problem Statement

Organizations increasingly rely on data-driven decision-making, yet a significant gap often exists between the theoretical data science knowledge gained in academic coursework and the practical, tool-based skills required to design and operate real-world data science pipelines. Students frequently lack structured, industry-mentored exposure to the complete lifecycle of a data science project — from identifying a suitable business problem, through data engineering and model building, to deployment and platform administration.

This internship was designed to address that gap by providing a structured, certification-driven curriculum that progressively builds competency from foundational business-use-case identification to advanced machine learning, deployment and enterprise platform administration, using an industry-standard tool (RapidMiner) rather than isolated academic exercises.

4.4 Project Objectives

- Identify real-world business problems suitable for Artificial Intelligence and Machine Learning solutions.
- Build strong data engineering skills — accessing, loading, transforming and cleansing data.
- Develop foundational and advanced Machine Learning capabilities using RapidMiner.
- Master model deployment, monitoring and complex predictive modeling techniques.
- Learn RapidMiner AI Hub administration and enterprise-level platform operations.
- Earn industry-recognized RapidMiner certifications at Professional and Master levels.

4.5 Scope of the Project

The scope of the internship was limited to the structured 8-week virtual curriculum defined by the AICTE–EduSkills National Internship Portal and Siemens Data Science Master track. It covered guided, exercise-based learning across seven modules — Applications & Use Cases, Data Engineering, Machine Learning, Data Preparation and

Platform Administration — at Professional and Master certification levels, assessed through weekly tests, project submissions and a Final Assessment Test.

The internship did not involve deployment to a live client environment or a paid engagement; it was unpaid and entirely virtual/self-paced within the weekly schedule. The scope was therefore focused on individual skill-building and certification rather than delivering a production system for an external client.

4.6 Technologies and Tools Used

Table 4.1 below summarizes the key technologies and tools used during the internship.

Category	Tool / Technology	Purpose
Core Platform	RapidMiner Studio	Visual workflow design for data preparation, modeling and evaluation
Platform Administration	RapidMiner AI Hub	Deployment, scheduling, monitoring and enterprise administration of models
Big Data Processing	RapidMiner Radoop	Execution of data workflows on big-data / Hadoop-based environments
Scripting & Automation	Regular Expressions, Macros, Scripting	Advanced text processing, automation and reusable process logic
Model Integration	R and Python Integration	Embedding custom R and Python scripts within RapidMiner processes
Connectivity	Web APIs	Connecting external data sources and services to RapidMiner workflows
Delivery Platform	AICTE National Internship Portal	Registration, curriculum delivery and credential validation

Table 4.1: Technologies and Tools Used

4.7 System Architecture

RapidMiner AI Hub follows a structured, end-to-end data science workflow that was used as the working architecture throughout the internship. Figure 4.1 illustrates this workflow, which was applied consistently across all module exercises: data is first accessed from various sources, prepared and cleansed, used to build predictive models, validated and optimized against benchmark metrics, and finally deployed and monitored through the AI Hub platform.

RapidMiner AI Hub — End-to-End Data Science Workflow



Figure 4.1: RapidMiner AI Hub – End-to-End Data Science Workflow

The Platform Administration Master module additionally covered the administrative layer surrounding this workflow — installation and configuration of RapidMiner Studio, management of the AI Hub server, Radoop integration for big-data execution, real-time scoring services, and the RapidMiner Marketplace — enabling the workflow above to be operated reliably at an enterprise scale.

4.8 Methodology

The internship followed the widely-used CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, which was formally introduced in the Applications & Use Cases Professional module and applied implicitly across subsequent modules. This methodology structures a data science project into business understanding, data understanding, data preparation, modeling, evaluation and deployment — closely mirroring the week-wise curriculum design of the internship itself.

Learning was delivered through a combination of guided instruction, hands-on RapidMiner exercises, weekly online assessments and project documentation submissions. Each of the eight weeks focused on a specific certification module, with the final week reserved for credential validation and an overall Internship Grade Point Assessment.

4.9 Expected Outcomes

- A working, practical understanding of the end-to-end data science lifecycle using an industry-standard platform.
- Seven RapidMiner (Altair) certifications spanning Applications & Use Cases, Data Engineering, Machine Learning and Platform Administration.
- Improved employability and confidence in applying AI/ML techniques to real business problems.

- A documented internship report, presentation and certification portfolio suitable for academic evaluation and future job applications.

4.10 Certificates of Completion and Other Communication / Mail Proof

This section contains the internship offer letter and all certificates of completion earned during the internship, provided as documentary proof of participation and successful completion.



Applications & Use Cases – Professional Certification



Applications & Use Cases – Master Certification



This is to certify that

Aditya Kumar

successfully passed the examination for

Data Engineering Professional Certification

The Professional level is the first of two certification levels and is rewarded for solid experience in the topic. Some of the topics tested at this level are:

Data Access, Basic Transformations, Working with Multiple Data Sets, Pivot Tables, Routines and Simple Text Processing.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service.


Knut Makowski
(Head of Education)

Data Engineering – Professional Certification



This is to certify that

Aditya Kumar

successfully passed the examination for

Data Engineering Master Certification

The Master level is the second level after Professional and stands for high proficiency in the topic. Some of the topics tested at this level are:

Loops and Branches, Advanced Text Processing, Exception Handling, Logging, Data Cleansing, Regular Expressions, Macros, Web APIs, and Scripting.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service


Knut Makowski
(Head of Education)

Data Engineering – Master Certification

This is to certify that

Aditya Kumar

successfully passed the examination for

Machine Learning Professional Certification

The Professional level is the first of two certification levels and is rewarded for solid experience in the topic. Some of the topics tested at this level are:

Classification, Regression, Scoring, Hold-out Validation, Correlations, Feature Importance, Clustering and Association Rules.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgespassport.com or another open badge service.


Knut Makowski
(Head of Education)

Machine Learning – Professional Certification

This is to certify that

Aditya Kumar

successfully passed the examination for

Machine Learning Master Certification

The Master level is the second level after Professional and stands for high proficiency in the topic. Some of the topics tested at this level are:

Complex Predictive Models, Cross Validation and Correct Validation, Parameter Optimization, Model Selection, Feature Engineering, Time Series and Forecasting and Integrating R and Python Models.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgespassport.com or another open badge service.


Knut Makowski
(Head of Education)

Machine Learning – Master Certification

This is to certify that

Aditya Kumar

successfully passed the examination for

Platform Administration Master Certification

The Master level stands for high proficiency in the topic. Some of the topics tested at this level are:

Platform Overview, RapidMiner Studio Installation, RapidMiner AI Hub Administration, RapidMiner Radoop, RapidMiner Real-Time Scoring, and RapidMiner Marketplace.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service.


Knut Makowski
(Head of Education)

Platform Administration – Master Certification

CHAPTER 5: BIBLIOGRAPHY/REFERENCES

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